

Weather and EPA AQI Dataset Summary

1. Pollutant data (EPA):

I downloaded daily **CO, NO₂, SO₂, PM2.5 and O₃** data for New York State, where each city has several monitoring stations.

a. City-day aggregation:

For each pollutant, I grouped by (date, city), took the **mean concentration** (typical city level) and the **maximum Daily AQI** (worst air in that city that day).

- i. mean to describe the city's "typical condition" (inputs)
- ii. max to define whether somewhere in the city air got bad (target)

b. Combine pollutants:

I merged all pollutant tables on (**date, city**) with an outer join, so any day where at least one pollutant is available is kept in the dataset.

2. Weather Data (Queens & Bronx):

I downloaded daily weather data for Queens and Bronx from Visual Crossing and

- a. I kept only the **temperature, humidity and wind_speed** column, and merged them with the pollutant data on (date, city) using an inner join (keeping only days with both pollution and weather).

3. Target label:

For each city-day, I took the maximum AQI across pollutants and used it to label each day as:

- i. 0 - 50 -> Healthy
- ii. ≥ 51 -> Unhealthy

- A) The Pollutant Datasets, I have downloaded from the link <https://www.epa.gov/outdoor-airquality-data/download-daily-data>.
- B) The Weather Data, I have downloaded from the link <https://www.visualcrossing.com/weather-data/>
- C) And the idea of labels, I got from the link <https://www.epa.gov/outdoor-air-qualitydata/air-data-aqi-plot>